

JACK BATES

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PROFILE

Final-year Business Information Systems student with a double degree from TBS Barcelona and hands-on experience shipping AI agents into live enterprise production. Builds deterministic-by-default systems: fixed logic for fixed work, models invoked only where judgement is required, validation kept outside the model.

EXPERIENCE

- EUROCONTROL — Power Platform & AI Solutions Engineer**
Skyline Traineeship. Designed and shipped a suite of AI agents for the Network Management Directorate, built on connectors and tools.
Brussels · Apr–Aug 2026
- Email triage & ticketing pipeline** — layered architecture over a shared service-desk mailbox: deterministic filters and request-type classification run first, an exception screen routes ambiguous mail to humans, and only clean cases reach the reasoning agent, which performs knowledge-base retrieval and returns a structured ticket / answer / route / discard decision. Evaluated over 25,000+ emails spanning ~15 months; measured ~40% of inbound volume as auto-ticketable and identified documentation coverage, not classification accuracy, as the binding constraint. Live in production.
 - Release Intelligence Agent** — ingests Microsoft 365 / Azure roadmap updates through a MCP server, assesses organisational impact against an internal knowledge base, and routes communications by priority band with human approval gating on high-impact items. Scoring is deliberately split from judgement: the model rates across seven weighted dimensions, deterministic code applies externally-editable weights to compute the band, so any decision can be reconstructed and retuned without re-prompting. 800+ updates processed; ~8 hrs/week of manual monitoring displaced.
 - ITSD Help-Desk Agent** — retrieval assistant for service-desk staff, integrated with ServiceNow. Effort concentrated on constraining assertion rather than expanding retrieval: scope validation rejects near-miss articles with a stated reason, historical cases never substitute for documented procedure, and unanswerable queries escalate rather than produce a plausible guess.
- Greenhill Fruit Farms — Roadside Seller.** Handled €1,000–€2,000 daily in cash, stock and customer operations. 2025

EDUCATION

- University of Galway — BSc Business Information Systems**
First Class Honours — Year 1 (71%), Year 2 (75%) · Academic Scholarship (Years 1 & 2)
2023–2027
- Notable modules:** Advanced Database Technologies, Cybersecurity, Data Networks & Communications, Information Systems Technology, Advanced Application Development.
- TBS Education, Barcelona — Bachelor of Management (Data Science for Business)**
Double-degree exchange year. Provisional grade: 17.90/ 20.
2025–2026
- Notable modules:** Programming, Big Data Modelling, Data Management & Visualization, The Business Aspect of AI, Econometrics, Data Governance.
- Coláiste Choilm — Leaving Certificate**
601 points

SELECTED PROJECTS

- Writing-Voice LoRA Fine-Tune** — fine-tuned Llama-3.2-1B-Instruct (rank-32 LoRA, Unsloth) on 51 hand-curated examples of my own writing; evaluated against the base model on three held-out prompts. Style transferred, facts did not — reported as a negative result with the failure modes documented. Code, dataset and adapter public.
- Healthcare Market Access Modelling (hackathon)** — built decision-tree classification and regression models over 10,000+ observations to analyse European reimbursement dynamics; applied multicollinearity diagnostics and class-imbalance handling.
- Personal Knowledge System** — built a personal knowledge agent (Obsidian + Claude Code) that turns raw notes and reading into a self-organising, interlinked wiki — auto-tagging, deduplicating and cross-linking entries.

SKILLS & CERTIFICATIONS

Technical: Python · SQL · R · Java · AI agent design (Copilot Studio, MCP connectors & tools) · LangGraph · Machine learning — supervised & unsupervised modelling (classification, regression, clustering) · MongoDB · Power BI

Professional: Collaboration · Adaptability · Self-direction · Stakeholder engagement

Certifications: ISO 8000 Master Data Quality Manager (ECCMA) · Enhance Agents with Autonomous Capabilities (Microsoft Applied Skills)

INTERESTS & ACTIVITIES

- Shamrocks GAA** — Club member, senior team.
- Tullamore RFC** — Leinster u18 Club Rugby Cup winner.
- Interests:** Philosophy, Cybernetics, History.

REFERENCES

Available upon request.